Part Name/Description	Unit Quantity	Unit Prototype Cost	Total Prototype Cost	Unit Production Cost	Total Production Cost	Manufacturer	Manufacturer Part #	Vendor Link
0.1 uF Ceramic Capacitor, ±10%, X7R, 50 V, radial THT	8	$0.23	$1.84	$0.23	$1.84	TDK	FG28X7R1H104KNT00	https://www.digikey.com/en/products/detail/tdk/FG28X7R1H104KNT00/732
10 uF Aluminum Electrolytic Capacitor, ±20%, 50 V, radial THT	3	$0.92	$2.76	$0.92	$2.76	Nichicon	UBW1H100MPD1TA	https://www.digikey.com/en/products/detail/nichicon/UBW1H100MPD1TA/
0.33 uF Ceramic Capacitor, ±10%, X7R, 50 V, radial THT	1	$0.40	$0.40	$0.40	$0.40	TDK	FA14X7R1H334KNU06	https://www.digikey.com/en/products/detail/tdk/FA14X7R1H334KNU06/586
1 uF Aluminum Electrolytic Capacitor, ±20%, 50 V, radial THT	1	$0.25	$0.25	$0.25	$0.25	Cornell Dubilier Knowles	105CKR050M	https://www.digikey.com/en/products/detail/cornell-dubilier-knowles/105CKR050M/5410527
470 uF Aluminum Electrolytic Capacitor, ±20%, 50 V, radial THT	1	$1.11	$1.11	$1.11	$1.11	Nichicon	UVK1H471MPD	https://www.digikey.com/en/products/detail/nichicon/UVK1H471MPD/2539
5 mm Red LED, through-hole	2	$0.28	$0.56	$0.14	$0.27	Kingbright	WP1503SURDK	https://www.digikey.com/en/products/detail/kingbright/WP1503SURDK/166
1N4148 Small-Signal Diode, DO-35 axial THT	4	$0.10	$0.40	$0.11	$0.44	Diotec Semiconductor	1N4148	https://www.digikey.com/en/products/detail/diotec-semiconductor/1N4148/21380193
5 mm Blue LED, through-hole	1	$0.47	$0.47	$0.47	$0.47	Kingbright	WP7113QBD/D	https://www.digikey.com/en/products/detail/kingbright/WP7113QBD-
10-bit SPI DAC, 8-PDIP through-hole	1	$1.97	$1.97	$1.97	$1.97	Microchip Technology	MCP4911-E/P	https://www.digikey.com/en/products/detail/microchip-
Fuse Block Cart 600V 15A PCB	1	$1.37	$1.37	$1.94	$1.94	Littelfuse Inc.	03540101ZXGY	https://www.digikey.com/en/products/detail/littelfuse-inc/03540101ZXGY/553974
750mA Fuse	2	$0.63	$0.63	$0.63	$1.26	Littelfuse Inc.	0312.750MXP	https://www.digikey.com/en/products/detail/littelfuse-inc/0312-750MXP/778266
4 Ohm 2 W Speaker, wire leads	1	$9.17	$9.17	$9.17	$9.17	Same Sky (formerly CUI Devices)	CDS-40304-L100	https://www.digikey.com/en/products/detail/same-sky-formerly-cui-devices/CDS-40304-L100/5355548
2-position 5.08 mm Screw Terminal Block, THT	1	$0.63	$0.63	$0.63	$0.63	Same Sky (formerly CUI Devices)	TB004-508-02BE	https://www.digikey.com/en/products/detail/same-sky-formerly-cui-devices/TB004-508-02BE/10064103
Switched DC Barrel Jack, 2.0 mm ID / 5.5 mm OD, THT	1	$0.70	$0.70	$0.75	$0.75	Same Sky (formerly CUI Devices)	PJ-102A	https://www.digikey.com/en/products/detail/same-sky-formerly-cui-devices/PJ-102A/275425
1x6 2.54 mm Male Header, through-hole	1	$0.13	$0.13	$0.12	$0.12	Sullins Connector Solutions	PRPC006SAAN-RC	https://www.digikey.com/en/products/detail/sullins-connector-solutions/PRPC006SAAN-RC/2775248
2x4 2.54 mm Male Header, through-hole	1	$0.46	$0.46	$0.17	$0.17	Sullins Connector Solutions	PRPC004DAAN-RC	https://www.digikey.com/en/products/detail/sullins-connector-solutions/PRPC004DAAN-RC/2775290
1x3 2.54 mm Jumper Header, through-hole	1	$0.10	$0.10	$0.10	$0.10	Sullins Connector Solutions	PRPC003SAAN-RC	https://www.digikey.com/en/products/detail/sullins-connector-solutions/PRPC003SAAN-RC/2775251
PNP Power Transistor, 80 V 1.5 A, TO-126/SOT-32 THT	2	$1.57	$3.14	$1.57	$3.14	STMicroelectronics	BD140-16	https://www.digikey.com/en/products/detail/stmicroelectronics/BD140-
NPN Power Transistor, 80 V 1.5 A, TO-126/SOT-32 THT	2	$1.44	$2.88	$1.44	$2.88	STMicroelectronics	BD139-16	https://www.digikey.com/en/products/detail/stmicroelectronics/BD139-
1 kOhm Resistor, ±5%, 1/4 W, axial THT	4	$0.10	$0.40	$0.10	$0.40	YAGEO	CFR-25JB-52-1K	https://www.digikey.com/en/products/detail/yageo/CFR-25JB-52-1K/96
10 kOhm Resistor, ±5%, 1/4 W, axial THT	3	$0.10	$0.30	$0.10	$0.30	YAGEO	CFR-25JB-52-10K	https://www.digikey.com/en/products/detail/yageo/CFR-25JB-52-10K/
100 kOhm Resistor, ±5%, 1/4 W, axial THT	1	$0.11	$0.11	$0.11	$0.11	YAGEO	CFR-25JB-52-100K	https://www.digikey.com/en/products/detail/yageo/CFR-25JB-52-100K/
2.2 kOhm Resistor, ±5%, 1/4 W, axial THT	2	$0.11	$0.22	$0.10	$0.20	YAGEO	CFR-25JB-52-2K2	https://www.digikey.com/en/products/detail/yageo/CFR-25JB-52-2K2/
100 Ohm Resistor, ±5%, 1/4 W, axial THT	2	$0.10	$0.20	$0.10	$0.20	YAGEO	CFR-25JB-52-100R	https://www.digikey.com/en/products/detail/yageo/CFR-25JB-52-100R/
SPST-NO Momentary Pushbutton, through-hole	1	$3.13	$3.13	$3.13	$3.13	C&K	KS11R22CQD	https://www.digikey.com/en/products/detail/c-k/KS11R22CQD/332670
PC Test Point, multipurpose, through-hole	7	$0.30	$2.10	$0.29	$2.03	Keystone Electronics	5010	https://www.digikey.com/en/products/detail/keystone-electronics/5010/255332
PIC18F57Q43 Curiosity Nano Development Board	1	$10.61	$10.61	$10.61	$10.61	Microchip Technology	DM164150	https://www.digikey.com/en/products/detail/microchip-
5 V Linear Regulator, 1.5 A, TO-220 through-hole	2	$1.08	$2.16	$0.62	$1.24	STMicroelectronics	L7805CV	https://www.digikey.com/en/products/detail/stmicroelectronics/L7805CV/5
Quad Rail-to-Rail Op Amp, 14-PDIP through-hole	2	$0.59	$1.18	$0.44	$0.88	Microchip Technology	MCP6004-I/P	https://www.digikey.com/en/products/detail/microchip-
TOTAL			**$49.38**		**$48.77**			

Datasheet Link	Supplier	Supplier Part #	Order Date	Order #	Received	Surplus	Schematic Reference Designators
https://product.tdk.com/en/system/files/dam/doc/product/capacitor/cera	DigiKey	445-FG28X7R1H104KNT00-ND					C1,C5,C6,C7,C9,C10,C14,C15
https://www.nichicon.co.jp/english/series_items/catalog_pdf/e-ubw.pdf	DigiKey	493-UBW1H100MPD1TA-ND					C2,C3,C4
https://product.tdk.com/system/files/dam/doc/product/capacitor/ceramic/	DigiKey	445-180408-1-ND					C8
https://www.cde.com/resources/catalogs/CKR_CKS.pdf	DigiKey	105CKR050M-ND					C12
https://www.nichicon.co.jp/english/products/pdfs/e-uvk.pdf	DigiKey	493-UVK1H471MPD-ND					C13
https://www.kingbrightusa.com/images/catalog/SPEC/WP1503SURDK.pdf	DigiKey	754-WP1503SURDK-ND					D1,D4
https://diotec.com/request/datasheet/1n4148.pdf	DigiKey	4878-1N4148CT-ND					D2,D3
https://www.kingbrightusa.com/images/catalog/SPEC/WP7113QBD-D.pdf	DigiKey	754-WP7113QBD/D-ND					D5
https://ww1.microchip.com/downloads/en/DeviceDoc/22248a.pdf	DigiKey	MCP4911-E/P-ND					DAC
https://www.littelfuse.com/assetdocs/354-omni-blok-series-datasheet?assetguid=6E94C133-AD48-47B7-8FFF-	DigiKey	354101-GY					F1
https://www.littelfuse.com/assetdocs/fuse-312-and-318-datasheet?assetguid=93D513CB-FAFC-4ADB-9054-	Digikey	F4782					F1
https://www.sameskydevices.com/product/resource/cds-40304-l100.pdf	DigiKey	102-3553-ND					Off-board speaker
https://www.sameskydevices.com/product/resource/tb004-508.pdf	DigiKey	102-6179-ND					J1
https://www.sameskydevices.com/product/resource/pj-102a.pdf	DigiKey	CP-102A-ND					J3
https://www.sullinscorp.com/catalogs/82_PAGE90-91_.100_MALE_HDR.pdf	DigiKey	35-PRPC006SAAN-RC-ND					J4
https://www.sullinscorp.com/catalogs/82_PAGE90-91_.100_MALE_HDR.pdf	DigiKey	35-PRPC004DAAN-RC-ND					J5
https://www.sullinscorp.com/catalogs/82_PAGE90-91_.100_MALE_HDR.pdf	DigiKey	35-PRPC003SAAN-RC-ND					JP2
https://www.st.com/resource/en/datasheet/bd140.pdf	DigiKey	497-14760-5-ND					Q1
https://www.st.com/resource/en/datasheet/bd139.pdf	DigiKey	497-12750-5-ND					Q2
https://www.yageo.com/upload/media/product/productsearch/datasheet/r	DigiKey	1.0KQBK-ND					R1,R2,R10,R11
https://www.yageo.com/upload/media/product/productsearch/datasheet/r	DigiKey	13-CFR-25JB-52-10K-ND					R3,R4,R12
https://www.yageo.com/upload/media/product/productsearch/datasheet/r	DigiKey	13-CFR-25JB-52-100K-ND					R5
https://www.yageo.com/upload/media/product/productsearch/datasheet/r	DigiKey	2.2KQBK-ND					R6,R7
https://www.yageo.com/upload/media/product/productsearch/datasheet/r	DigiKey	100QBK-ND					R8,R9
https://www.ckswitches.com/media/1467/ks.pdf	DigiKey	CKN4080-ND					SW3
https://www.keyelco.com/product.cfm/product_id/274	DigiKey	36-5010-ND					TP1,TP2,TP3,TP4,TP5,TP6,TP7
https://ww1.microchip.com/downloads/en/DeviceDoc/PIC18F57Q43-Curiosity-	DigiKey	150-DM164150-ND					U1
https://www.st.com/resource/en/datasheet/l7805.pdf	DigiKey	497-1443-5-ND					U3
https://ww1.microchip.com/downloads/en/DeviceDoc/21733j.pdf	DigiKey	MCP6004-I/P-ND					U2A-U2E